

Towards Cross-Modal Preservation, Consistency and Alignment for Privacy-Preserving Visible-Infrared Person Re-Identification

Yudi Xie, Zhongao Zhou, Bin Yang[†], Zhenghan Chen, Mang Ye[†]
 National Engineering Research Center for Multimedia Software,
 School of Computer Science, Wuhan University.

{xieyudi, yangbin_cv, yemang}@whu.edu.cn

https://github.com/Dige945/PPA_CVPR26

Abstract

*Privacy-preserving Person Re-Identification (PP-ReID) addresses the core privacy-utility trade-off in Re-ID by retrieving a person across multiple non-overlapping cameras while applying anonymization techniques to protect sensitive information. However, prior PP-ReID studies are confined to single-modality visible scenarios, as 24-hour surveillance systems require robust cross-modal visible-infrared (VI) capabilities. Extending PP-ReID to the cross-modal VI setting is therefore crucial. Accordingly, we introduce a new task: Privacy-Preserving Visible-Infrared Person Re-Identification (PP-VI-ReID). This task presents two severe challenges: 1) Crude anonymization strategies destroy identity-critical information and disrupt cross-modal alignment 2) The anonymization process creates inconsistent distortions across modalities. It disrupts color-based textures in visible images while obscuring thermal contours in infrared images. This inconsistency with modality gap forms a Mixed Gap. To overcome these challenges, we propose a framework, the **Precise Privacy-preserving and Alignment Network (PPA)** with two components: 1) A Keypoint-Preserving Regularization (KPR) module leverages human pose as a prior to guide structure-aware anonymization, preserving essential body features. 2) A Differential Consistency-guided Modality Alignment (DCMA) treats anonymization perturbations not as varying noise, but as a stable, learnable offset, facilitating robust alignment between raw and anonymized features across modalities. Experiments on SYSU-MM01 and RegDB validate our framework, establishing a strong baseline.*

1. Introduction

Person re-identification (ReID) aims at matching the same person image captured by non-overlapping cameras [31,

[†]Corresponding Author

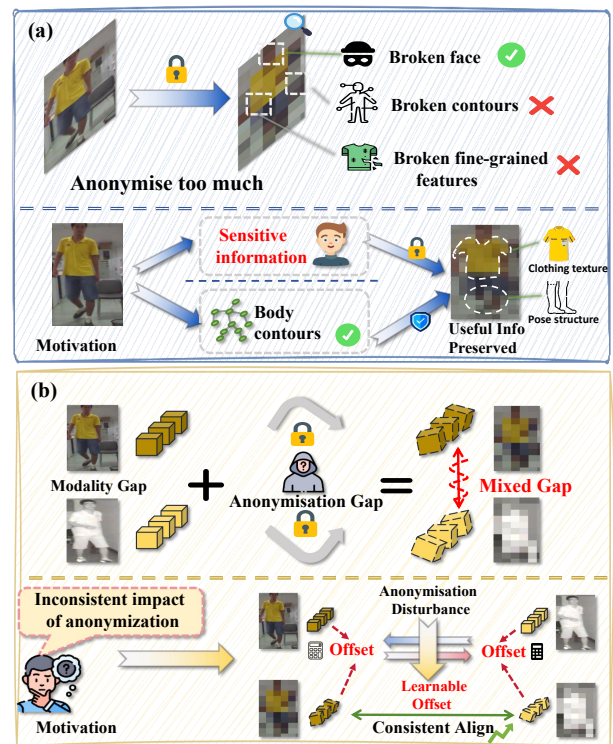


Figure 1. Challenges and motivation. **(a) Anonymization Challenge:** “One-size-fits-all” anonymization (Anonymise too much) excessively breaks critical fine-grained features and contours, causing a loss of identity information. **Motivation:** To preserve these features for effective recognition while protecting sensitive information. **(b) Feature-Align Challenge:** The Modality Gap and Anonymisation Gap (causing inconsistent distortions) combine into a mixed gap, leading to feature alignment failure. **Motivation:** To convert these inconsistent distortions into a consistent, learnable offset, enabling robust alignment of feature differences across modalities.

33, 34, 42], but large-scale pedestrian imagery capturing sensitive personal information like appearance and body

shape poses significant privacy risks [5, 9, 12, 13, 24, 25, 30]. To address this, the Privacy-preserving Person Re-Identification (PP-ReID) task is introduced by prior work SecureReID [44], which uses anonymization methods (e.g., GANs) and trains robust ReID models across raw and anonymized domains to enable cross-domain retrieval without performance compromise.

In practical applications, visible-light cameras are sensitive to illumination variations and often fail to capture reliable information under low-light conditions [8, 14]. Consequently, infrared cameras are widely deployed during nighttime or low-visibility situations. This raises the Visible-Infrared Person Re-Identification (VI-ReID) task, which aims to match pedestrians across these modalities for 24-hour surveillance. However, these VI-ReID studies [8, 33, 36, 50] focus on retrieval accuracy and largely overlook the critical privacy risks. To address this, we propose the Privacy-Preserving Visible-Infrared Person Re-Identification (**PP-VI-ReID**) task, which generates anonymized visible-infrared images to preserve privacy while enabling high-performance cross-modal retrieval.

Current privacy-preserving frameworks [1, 4, 15, 16, 19, 22, 23, 44, 47, 49] are ill-equipped for this task as their crude, “one-size-fits-all” anonymization strategies. While these methods conceal appearance details, they inevitably inflict destructive loss on fine-grained features and structural information (such as body contours). This loss of identity-critical information severely degrades ReID performance. This raises a pivotal problem: ***I) how to transition from crude anonymization to a precise, structure-aware approach that preserves the key structural and fine-grained features needed for matching?***

In contrast to single-modality scenarios, this degradation is exacerbated by the asymmetric nature of the anonymization impact across modalities. Initial attempts to align features using standard shared-space methods [42, 43] are rendered ineffective. These methods are designed to learn invariants from a stable modality gap. However, they collapse because anonymization distorts color textures in visible images while blurring thermal contours in infrared images. These failures reveal a critical gap: existing cross-modal techniques are incapable of learning invariant representations once the anonymization perturbations distort the feature space. This raises another pivotal problem: ***II) how to bridge the mixed gap, where the stable modality gap is compounded by asymmetric anonymization distortions?***

To address above problem, we propose the “Precise Privacy-preserving and Alignment Network” (PPA), a novel cross-modal privacy-preserving Re-ID framework with two core modules. For problem ***I)***, we propose **Keypoint-Preserving Regularization(KPR)** to resolve the issue of excessive information destruction. It acts as a crucial constraint on the destructive extent of the transforma-

tion. By incorporating human pose information as prior knowledge, it compels the generator to find an optimal balance between two competing objectives: achieving robust privacy protection and preserving the structural integrity of identity-critical keypoint regions. This forces the generator to learn an intelligent anonymization that selectively alters identity cues while retaining the essential pose structure required for effective re-identification. For problem ***II)***, we propose **Differential Consistency-guided Modality Alignment (DCMA)**. This module enforces consistency between feature difference vectors of original-anonymized pairs across modalities. It treats the typically inconsistent anonymization perturbations as a consistent, learnable offset, ensuring predictable semantic impact and robust alignment despite feature space distortion.

The main contributions of this paper are as follows:

- **Task Contribution.** We introduce Privacy-Preserving Visible-Infrared Person Re-Identification (PP-VI-ReID), a new task focusing on generating anonymized VI images to protect privacy while enabling high-performance cross-modal retrieval.
- **Methodological Contribution.** We propose an innovative framework, the Precise Privacy-Preserving and Alignment Network (PPA), which includes a KPR module to maximize the retention of identity information while protecting privacy, and a DCMA module to resolve the alignment challenges caused by the coupling of anonymization and modality gaps.
- **Empirical Contribution.** Extensive experiments on SYSU-MM01 and RegDB demonstrate PPA’s outstanding performance and robustness across various privacy-preserving cross-modal ReID scenarios, establishing a strong baseline for PP-VI-ReID tasks.

2. Related work

2.1. Single-Modality Privacy-Preserving Person ReID

Raw surveillance images often capture pedestrians with sensitive private information, prompting public concern and the subsequent removal of certain datasets, which hampers progress in pedestrian re-identification (ReID) and related recognition tasks [2, 45]. Early ReID research [17, 52] concentrate on boosting recognition accuracy and enhancing cross-domain adaptability. However, as privacy issues have taken center stage, conventional anonymization techniques—such as blurring or pixelation—have revealed a trade-off: complete de-identification frequently results in a significant loss of utility.

Recent approaches have sought to mitigate this issue. For instance, PIS [4] leverages the interpolation capabilities of variational autoencoders (VAEs) to “translate” pedestrian images, thereby removing absolute identity information

while preserving relative identity relationships. Similarly, AVIH [23] iteratively diminishes the association between protected and raw images to safeguard privacy without compromising their proximity in feature space. Moreover, PixelFade [49] introduces a noise-guided objective coupled with a heuristic optimization algorithm—alternating between constraint and partial replacement operations—to incrementally “noise” the images, effectively masking sensitive details while retaining discriminative features for ReID.

Under cross-modal ReID scenarios, however, the accuracy of these methods deteriorates significantly. In contrast, our proposed PPA markedly improves cross-modal retrieval performance while maintaining robust privacy protection.

2.2. Visible-Infrared Person ReID

Visible-infrared person re-identification (VI-ReID)[32, 36, 38, 42, 46] presents a formidable challenge in cross-modal recognition, aiming to enhance query accuracy by bridging the significant modality gap between visible and infrared images of the same identity [8, 18, 21, 29, 31, 33, 35, 37, 39–41, 53, 54]. The pioneering work in Deep Zero-Padding [28] initiates this domain by employing grayscale images to train and test a zero-filled single-stream network. Later, XIV [14] introduces an auxiliary X-modality, leveraging a shared-weight feature extractor to process three modalities concurrently, thereby alleviating the disparity between infrared and visible data. Furthermore, CMT [11] integrates Transformers into cross-modal re-identification, exploiting their capacity to capture global relationships and enhance feature complementarity through specialized modality embeddings. In parallel, MSPAC [48] devises a multi-scale cascade framework for progressively aggregating fine-grained features, while CAJ [43] proposes a channel enhancement method that generates color-independent images by randomly exchanging color channels in VI-ReID.

Despite these advancements, prior studies largely neglect privacy protection, leading to inadequate accuracy in cross-modal retrieval using anonymized images. Our work addresses this critical gap by exploring privacy-preserving cross-modal ReID methods, a development that holds significant promise for the real-world deployment of VI-ReID.

3. Proposed Method

Our framework, the Precise Privacy-Preserving and Alignment Network (PPA), is illustrated in Fig. 2. It incorporates two key components: Keypoint-Preserving Regularization (KPR) and Differential Consistency-guided Modality Alignment (DCMA). PPA utilizes a UNet-based dual-stream generator to create anonymized versions of visible and infrared images. Subsequently, both original and anonymized image pairs are processed by a shared feature extraction backbone. To achieve precise anonymization, the KPR module leverages a pre-trained pose model

to guide a mask head, focusing the anonymization on identity-sensitive keypoint regions. For robust alignment, the DCMA module minimizes the distance between the feature difference vectors (raw vs. anonymized) across modalities, ensuring a consistent transformation. The synergy of KPR’s precision and DCMA’s alignment enables PPA to learn discriminative yet privacy-protected representations for cross-modality re-identification.

To facilitate the description of our method, the notations used in this paper are first introduced. Let $X_v = \{x_1^v, x_2^v, \dots, x_N^v\}$ denote the visible images with N samples and $X_i = \{x_1^i, x_2^i, \dots, x_M^i\}$ denote the infrared images with M samples. $Y_v = \{y_1^v, y_2^v, \dots, y_N^v\}$ and $Y_i = \{y_1^i, y_2^i, \dots, y_M^i\}$ denote the anonymized visible and infrared images generated by our dual-stream generator. Additionally, $F_v = \{f_1^v, f_2^v, \dots, f_N^v\}$ and $F_{va} = \{f_1^{va}, f_2^{va}, \dots, f_N^{va}\}$ represent visible raw and anonymized features extracted by the visible feature extractor f_θ^v , similarly $F_i = \{f_1^i, f_2^i, \dots, f_M^i\}$ and $F_{ia} = \{f_1^{ia}, f_2^{ia}, \dots, f_M^{ia}\}$ represent infrared raw and anonymized features extracted by the infrared feature extractor f_θ^i .

3.1. Overall Architecture

Generator Network. Our framework utilizes a dual-stream generator based on the UNet architecture to produce the anonymized images.

The anonymization process is driven by an adversarial training scheme. Focusing on the visible modality, the generator is trained adversarially against a corresponding discriminator D_Y^{rgb} . The generator’s objective is to produce an anonymized image y^v that is indistinguishable from a target anonymized image $\hat{y}^v$. This target, $\hat{y}^v$, is created by applying a traditional privacy-preserving method (e.g., mosaic, blurring, or adding noise) directly to the original input image x_v . This adversarial dynamic pushes the generator to create outputs that are both realistic and adhere to a privacy-preserving style. The infrared stream operates analogously, employing its own discriminator D_Y^{ir} and a similarly generated target $\hat{y}^i$ to ensure the generation of high-quality anonymized infrared images y_i .

Mathematically, the adversarial objective function for the generator is defined as:

$$\begin{aligned} \mathcal{L}_{adv} = & \frac{1}{N} \sum_{j=1}^N (\log(1 - D_Y^{rgb}(x_j^v, y_j^v)) + \log D_Y^{rgb}(x_j^v, \hat{y}_j^v)) \\ & + \frac{1}{M} \sum_{k=1}^M (\log(1 - D_Y^{ir}(x_k^i, y_k^i)) + \log D_Y^{ir}(x_k^i, \hat{y}_k^i)). \end{aligned} \quad (1)$$

Besides the adversarial objective, a reconstruction loss $\mathcal{L}_{rec}$ is introduced to enforce pixel-level consistency between the generated images and their targets. This loss

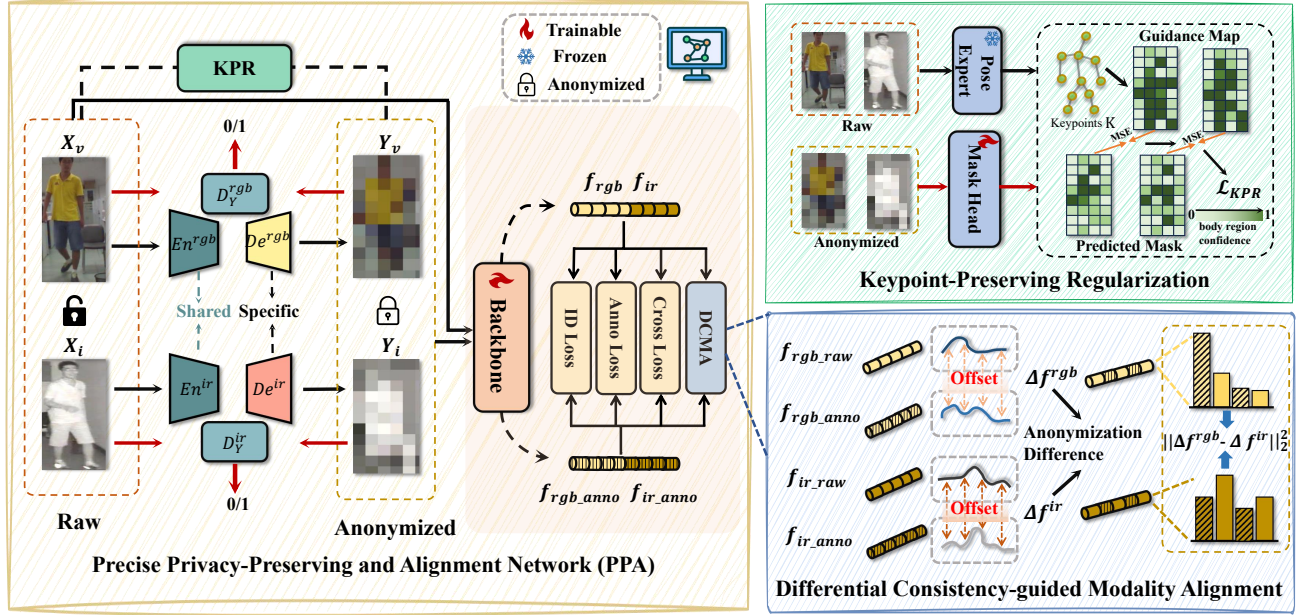


Figure 2. Illustration of our Precise Privacy-Preserving and Alignment Network (PPA) framework. Our model enhances a baseline (left), which includes a dual-stream generator and a ReID backbone, with two key components. The Keypoint-Preserving Regularization (KPR) module (top-right) uses keypoint masks to guide the generator, achieving precision anonymization that preserves identity features while protecting sensitive information. Concurrently, the Differential Consistency-guided Modality Alignment (DCMA) module (bottom-right) aligns the feature-level differences between raw and anonymized images, ensuring robust cross-modal consistency.

guarantees that the mapping from an individual input, x^v or x^i , to its corresponding output, y^v or y^i , closely matches the target anonymized image, $\hat{y}^v$ or $\hat{y}^i$. Calculated using the $\mathcal{L}_1$ norm, the loss is formally defined as follows:

$$\mathcal{L}_{rec} = \frac{1}{N} \sum_{j=1}^N \|\hat{y}_j^v - y_j^v\|_1 + \frac{1}{M} \sum_{k=1}^M \|\hat{y}_k^i - y_k^i\|_1. \quad (2)$$

The total loss of our dual-stream generator is:

$$\mathcal{L}_{gen} = \mathcal{L}_{adv} + \lambda \mathcal{L}_{rec}. \quad (3)$$

where λ is a hyperparameter to reduce the artifacts [10].

Backbone Network. Our baseline employs a two-stream architecture with a visible feature extractor f_{θ}^v and an infrared feature extractor f_{θ}^i . During training, the original images X_v and X_i are fed into their respective extractors to obtain feature sets F_v and F_i . Similarly, the anonymized images Y_v and Y_i are processed to produce their corresponding anonymized feature sets F_{va} and F_{ia} .

To mitigate the significant modality gap, a knowledge distillation strategy is also employed from a pre-trained cross-modal model (the “teacher”) to our feature extractors (the “student”). Specifically, our teacher model employs AGW model [42] pre-trained on cross-modal image

datasets, while the student model utilizes the same backbone. The original images X_v and X_i are passed through the frozen teacher model to obtain expert features f_j^{tv} and f_j^{ti} . The student is then trained to mimic the teacher by minimizing the Cross-modal robust distillation loss, which encourages the learning of modality-invariant features:

$$\mathcal{L}_{cross} = \sum_{j=1}^N \|f_j^{tv} - f_j^v\|_2^2 + \sum_{k=1}^M \|f_k^{ti} - f_k^i\|_2^2, \quad (4)$$

where N and M denote the number of training samples in the visible and infrared modalities and $\|\cdot\|_2^2$ represents $\mathcal{L}_2$ norm, respectively.

To bridge the domain gap and enable retrieval between original and anonymized images, a feature alignment loss $\mathcal{L}_{anno}$ is introduced. This objective minimizes the feature distance between an original image and its anonymized version within each modality by computing the Mean Squared Error (MSE) as follows:

$$\mathcal{L}_{anno} = \frac{1}{B} \sum_{j=1}^B (\|f_j^v - f_j^{va}\|_2^2 + \|f_j^i - f_j^{ia}\|_2^2), \quad (5)$$

where B is the batch size, and $f_j^v, f_j^{va}, f_j^i, f_j^{ia}$ correspond to the features of the j -th instance in a batch.

The total objective for the backbone network $\mathcal{L}_{reid}$, integrates these components with a standard identification objective to ensure feature discriminability. This objective consists of a Cross-Entropy Loss $\mathcal{L}_{ce}$ [26] and a Triplet Loss $\mathcal{L}_{tri}$ [7]. The total loss of backbone network is:

$$\mathcal{L}_{reid} = \mathcal{L}_{ce} + \mathcal{L}_{tri} + \beta \mathcal{L}_{cross} + \gamma \mathcal{L}_{anno}, \quad (6)$$

where β and γ are hyperparameters that balance the contributions of the respective objectives.

Finally, we obtain the baseline objective $\mathcal{L}_b$:

$$\mathcal{L}_b = \mathcal{L}_{gen} + \mathcal{L}_{reid}. \quad (7)$$

3.2. Keypoint-Preserving Regularization

To ensure that the anonymization process specifically targets identity-rich regions while preserving other useful information (such as contours), we introduce the Keypoint-Preserving Regularization (KPR) module. Instead of applying anonymization uniformly, KPR functions as a structural regularizer, learns to generate a dynamic mask that precisely localizes human body keypoints, which are highly correlated with identity. This regularization mechanism leverages an expert, pre-trained pose estimation model to supervise the structural integrity of the generator’s output.

The KPR module consists of two components: a pose-based guidance map generator and a learnable mask head.

Pose-based Guidance Map Generation. A pre-trained YOLOv8n-pose model is employed as a frozen “expert” to detect human keypoints in the original input images X_v and X_i . For each image, the model identifies the coordinates of various body joints, identifies a set of keypoint coordinates $K = \{k_1, \dots, k_N\}$ for the most prominent person, where each keypoint k_j is located at pixel (u_j, v_j) . A guidance map $C_{pose} \in \{0, 1\}^{H \times W}$ is then constructed. Its value at each pixel (u, v) is determined by whether the pixel lies within a predefined radius R of any detected keypoint. This is formally defined as:

$$C_{pose}(u, v) = \begin{cases} 1 & \text{if } \min_j \sqrt{(u - u_j)^2 + (v - v_j)^2} \leq R, \\ 0 & \text{otherwise} \end{cases}, \quad (8)$$

where R is a predefined radius. This equation states that a pixel’s value is 1 if its distance to the nearest keypoint is within the radius R , and 0 otherwise. This process effectively creates a binary mask by taking the union of all circular regions around the keypoints, serving as a stable and accurate supervision signal.

Learnable Mask Head. We design a lightweight convolutional network, termed “mask head” (m_ϕ), to predict these keypoint regions directly from the anonymized images Y_v and Y_i . We use separate but identical mask heads for each modality, m_ϕ^v and m_ϕ^i . Each head consists of a

sequence of 3×3 convolutional layers followed by a final Sigmoid activation function. It takes an anonymized image $Y \in \mathbb{R}^{H \times W \times 3}$ as input and outputs a single-channel soft mask $M_{pred} \in [0, 1]^{H \times W}$. This process is defined as:

$$M_{pred}^v = m_\phi^v(Y_v) \quad \text{and} \quad M_{pred}^i = m_\phi^i(Y_i). \quad (9)$$

Pose Guidance Loss. To train the mask heads, the process enforces that their predicted masks M_{pred} closely match the guidance maps C_{pose} generated by the pose model. This is achieved by minimizing a pose guidance loss $\mathcal{L}_{KPR}$, defined as the mean squared error between the two masks. By optimizing this loss, the mask heads learn to accurately identify keypoint regions even in the anonymized domain. The loss is formulated as:

$$\mathcal{L}_{KPR} = \frac{1}{B} \sum_{j=1}^B (\|M_{pred,j}^v - C_{pose,j}^v\|_2^2 + \|M_{pred,j}^i - C_{pose,j}^i\|_2^2). \quad (10)$$

3.3. Differential Consistency-guided Modality Alignment

A primary challenge in cross-modal Re-ID is the significant domain gap between visible and infrared images. The anonymization process, which acts as a domain transformation, can inadvertently exacerbate this gap. This inconsistent transformation can disrupt the feature alignment necessary for accurate cross-modal matching.

To address this, we propose the Differential Consistency-guided Modality Alignment (DCMA) module. Instead of only aligning the final feature representations, DCMA enforces consistency on the transformation caused by the anonymization itself. The core intuition is that the change or shift in the feature space resulting from anonymization should be similar for both modalities.

To quantify this change, we define the “anonymization difference vector” for each modality. For a given person, this vector is the difference between the feature of the original image and the feature of its anonymized version. Using the notations defined previously, the difference vectors for the visible and infrared modalities are:

$$\Delta f^v = f^v - f^{va}, \quad (11)$$

$$\Delta f^i = f^i - f^{ia}, \quad (12)$$

These vectors, Δf^v and Δf^i , capture the semantic displacement in the feature space induced by the anonymization process for each modality.

The DCMA module aims to align these two difference vectors. By forcing the anonymization-induced changes to be parallel and have similar magnitudes for both modalities, we ensure the generator and feature extractors learn a

consistent cross-modal anonymization policy. The DCMA objective can be formulated as:

$$\begin{aligned}\mathcal{L}_{DCMA} &= \frac{1}{B} \sum_{j=1}^B \|\Delta f_j^v - \Delta f_j^i\|_2^2 \\ &= \frac{1}{B} \sum_{j=1}^B \|(f_j^v - f_j^{va}) - (f_j^i - f_j^{ia})\|_2^2.\end{aligned}\quad (13)$$

Finally, all components are integrated to formulate a comprehensive optimization objective:

$$\mathcal{L}_{PPA} = \mathcal{L}_b + \delta \mathcal{L}_{KPR} + \eta \mathcal{L}_{DCMA}. \quad (14)$$

where δ and η are hyperparameters that balance the contribution of each term.

Discussion. To theoretically justify the necessity of DCMA, we provide a proof based on the Triangle Inequality. The cross-modal alignment error in the anonymized space is bounded by:

$$\begin{aligned}\|f^{va} - f^{ia}\| &= \|(f^v - f^i) - (\Delta f^v - \Delta f^i)\| \\ &\leq \|f^v - f^i\| + \|\Delta f^v - \Delta f^i\|.\end{aligned}\quad (15)$$

Without DCMA, even with perfectly aligned raw features ($f^v \approx f^i$), inconsistent shifts ($\Delta f^v \neq \Delta f^i$) introduce an irreducible error term. DCMA minimizes this exact term $\|\Delta f^v - \Delta f^i\|$, proving mathematically essential for cross-modal consistency. In contrast to prior PP-ReID works [16, 44], this formalizes treating inconsistent distortion as a learnable offset for robust alignment.

4. Experiment

4.1. Datasets and Evaluation Metrics

The proposed approach is evaluated on two public cross-modal datasets: SYSU-MM01 [28] and RegDB [20]. SYSU-MM01 contains 395 training identities with images from 2 near-infrared and 4 visible cameras, evaluated under all-search and indoor-search modes. RegDB contains 412 identities from a dual-camera (visible/thermal) system. Results are averaged over 10 random splits.

The evaluation covers two aspects: anonymization quality and ReID performance. Anonymization quality is assessed by the Structural Similarity Index (SSIM) for perceptual similarity and the Peak Signal-to-Noise Ratio (PSNR) for distortion [27]. $SSIM < 0.5$ and $PSNR < 15$ are considered thresholds for making images visually indistinguishable. ReID performance is measured by rank-k matching accuracy, mean average precision (mAP) [51], and mean inverse negative penalty (mINP) [43].

4.2. Implementation Details

The proposed framework is implemented in PyTorch. The ReID model incorporates two shallow feature extractors

Table 1. **Comparison with SecureReID methods on SYSU-MM01.** For infrared images as query sets and visible images as gallery sets, we set four test conditions for comparison, including Raw to Raw, Raw to Anon, Anon to Raw and Anon to Anon. Rank1 accuracy(%), mAP (%) and mINP (%) are reported.

SYSU	query		gallery		All search			Indoor search		
	Raw	Anon	Raw	Anon	Rank1	mAP	mINP	Rank1	mAP	mINP
<i>Initialized by mosaic.</i>										
SecureReID	✓		✓		35.6	37.3	25.0	46.0	56.2	51.9
	✓			✓	31.9	31.0	17.3	45.2	53.7	47.3
		✓	✓		31.7	29.5	16.0	36.9	48.3	44.8
		✓		✓	32.2	30.4	16.8	31.0	42.2	37.6
ours	✓		✓		60.7	58.8	45.0	70.3	75.6	72.2
	✓			✓	50.9	46.4	29.4	61.9	68.7	64.4
		✓	✓		53.3	51.9	37.9	62.9	70.6	67.6
		✓		✓	46.2	43.4	27.8	56.6	65.1	61.0
<i>Initialized by blurring.</i>										
SecureReID	✓		✓		29.8	31.2	18.8	36.5	48.3	43.8
	✓			✓	30.7	31.0	18.2	41.0	50.7	45.2
		✓	✓		23.4	25.9	15.1	31.7	42.8	38.4
		✓		✓	20.7	22.7	12.6	29.1	40.6	36.2
ours	✓		✓		52.6	50.4	35.2	69.1	74.5	70.9
	✓			✓	47.6	43.9	28.8	62.2	67.9	63.2
		✓	✓		46.3	44.7	30.6	58.2	66.9	63.7
		✓		✓	44.3	41.0	26.1	51.6	60.5	56.3
<i>Initialized by noise.</i>										
SecureReID	✓		✓		41.7	41.2	27.6	50.3	59.9	55.6
	✓			✓	33.9	35.5	22.4	46.2	56.7	51.7
		✓	✓		34.4	35.3	23.2	44.7	55.6	51.5
		✓		✓	29.1	31.8	19.7	40.4	51.4	46.5
ours	✓		✓		59.0	56.1	40.0	59.1	67.2	63.9
	✓			✓	50.0	44.5	26.5	51.5	60.4	56.0
		✓	✓		47.2	46.3	30.9	52.4	62.3	59.1
		✓		✓	41.7	39.5	23.5	47.3	57.6	53.4

configured as in AGW [42], while a shared feature extractor based on ResNet50 [6] is initialized with ImageNet pre-trained weights [3]. Moreover, the supervision image is initialized using three algorithms—mosaic, blur and Gaussian noise—with the mosaic radius set to 24, the blur radius to 12, and the Gaussian noise variance to 0.5.

During training, the model runs for 120 epochs and evaluates privacy quality and ReID performance every 3 epochs. In each training batch, 8 identities are randomly selected with 4 instances per modality for each identity, and resize input images to 256×128 . We utilize the SGD optimizer with a momentum of 0.9. For our dual-stream generator, the learning rate increases linearly from 3.5×10^{-5} to 3.5×10^{-4} during first 10 epochs, decreases to 3.5×10^{-5} by epoch 40, and further decays to 3.5×10^{-6} by epoch 80. For consistency, the ReID model’s learning rate follows the scheduling detailed in AGW. Additionally, the hyperparameters are set as follows: $\lambda = 100$, $\beta = 0.5$, $\gamma = 1.0$, $\delta = 1.0$, $\eta = 0.1$ and $R = 5$. This specific configuration was chosen to balance the contributions of the individual components, ensuring that the training process effectively addresses both the modality and anonymization gaps inherent in the PP-VI-ReID task.

Table 2. **Comparison with SecureReID methods on SYSU-MM01 and RegDB.** We evaluate the quality of the generated anonymized visible images and infrared images respectively. SSIM and PSNR are reported.

	Methods	SYSU-MM01 (all search)		SYSU-MM01 (indoor search)		RegDB	
		RGB	IR	RGB	IR	RGB	IR
<i>Initialized by mosaic.</i>							
SSIM	SecureReID ours	0.13	0.30	0.15	0.30	0.14	0.29
		0.08	0.18	0.07	0.15	0.33	0.16
PSNR	SecureReID ours	7.8	10.3	8.1	10.3	8.3	6.6
		6.9	10.1	7.1	10.2	6.6	9.2
<i>Initialized by blurring.</i>							
SSIM	SecureReID ours	0.16	0.33	0.18	0.32	0.16	0.43
		0.11	0.25	0.15	0.23	0.32	0.20
PSNR	SecureReID ours	9.5	11.5	10.1	11.5	9.3	7.4
		8.2	10.5	9.2	10.3	6.3	10.4
<i>Initialized by noise.</i>							
SSIM	SecureReID ours	0.07	0.26	0.07	0.26	0.09	0.31
		0.03	0.14	0.03	0.15	0.15	0.04
PSNR	SecureReID ours	7.5	9.3	7.4	9.3	7.8	6.7
		5.8	8.3	5.3	7.7	6.3	7.1

4.3. Results of Visible-Infrared Person Re-ID

As the first work to address privacy protection in visible-infrared queries, we compare our PPA framework against SecureReID[44], a state-of-the-art method for single-modality PP-ReID. For a fair comparison on our cross-modal PP-VI-ReID task, SecureReID is adapted and trained using AGW and a single-stream adversarial network.

For PP-VI-ReID, infrared images are used as queries and visible images as the gallery, evaluating performance across four scenarios: **Raw to Raw** (raw query/gallery), **Raw to Anon** (raw query, anonymized gallery), **Anon to Raw** (anonymized query, raw gallery), and **Anon to Anon** (both query and gallery anonymized).

As shown in Table 1 and Table 3, our proposed PPA framework outperforms the SecureReID across all test settings. Specifically, for the Raw to Raw task with mosaic-initialized supervision images, Rank1 accuracy improves by 25.1% on SYSU-MM01(all search) and 34.6% on RegDB(infrared to visible), demonstrating our PPA’s ability to mitigate the adverse effects of anonymization on ReID performance. Even under the most challenging Anon to Anon scenario, improvements of 14.0% and 11.4% in Rank1 accuracy on SYSU-MM01(all search) and RegDB(infrared to visible), respectively, highlight the strong cross-modal retrieval capabilities of our approach in the presence of both modality and anonymization gaps.

4.4. Results of Visible-Infrared Privacy Protection

To assess the quality of anonymized images generated by our PPA framework, we evaluate the PSNR and SSIM metrics of both visible and infrared anonymized images relative to their raw versions on the SYSU-MM01 and RegDB datasets, comparing them against the SecureReID. Table 2

Table 3. **Comparison with SecureReID methods on RegDB.** We set four test conditions for comparison, including Raw to Raw, Raw to Anon, Anon to Raw and Anon to Anon. Rank1 accuracy(%), mAP (%) and mINP (%) are reported.

RegDB	query		gallery		Infrared to Visible			Visible to Infrared		
	Raw	Anon	Raw	Anon	Rank1	mAP	mINP	Rank1	mAP	mINP
<i>Initialized by mosaic.</i>										
SecureReID	✓		✓		45.9	45.3	34.4	52.4	51.7	39.2
	✓		✓	✓	36.7	36.7	26.0	36.6	37.3	26.2
		✓	✓		27.3	28.7	20.2	32.0	34.0	25.8
		✓	✓	✓	28.3	29.5	20.7	31.1	32.0	22.8
ours	✓		✓		80.5	74.6	59.0	81.0	76.0	62.4
	✓		✓	✓	61.3	55.0	37.8	45.7	40.8	25.5
		✓	✓		32.3	32.5	22.8	51.3	49.1	37.4
		✓	✓	✓	39.7	36.0	22.3	41.7	37.5	23.1
<i>Initialized by blurring.</i>										
SecureReID	✓		✓		46.7	46.6	34.4	45.8	46.3	36.5
	✓		✓	✓	33.9	34.5	24.7	36.3	37.6	28.8
		✓	✓		34.1	34.6	23.5	32.0	34.4	26.2
		✓	✓	✓	42.7	39.7	28.4	41.8	40.9	29.8
ours	✓		✓		80.0	74.0	58.5	80.8	75.4	61.8
	✓		✓	✓	62.3	56.3	39.3	46.6	41.9	26.5
		✓	✓		35.0	34.6	24.0	53.2	50.8	38.6
		✓	✓	✓	41.3	37.5	23.6	45.3	39.8	24.5
<i>Initialized by noise.</i>										
SecureReID	✓		✓		35.8	37.2	26.6	42.0	43.5	34.5
	✓		✓	✓	34.8	34.5	24.0	38.5	38.9	28.0
		✓	✓		30.8	31.8	21.7	33.5	35.1	26.8
		✓	✓	✓	39.5	35.8	22.5	36.8	35.6	24.3
ours	✓		✓		80.8	74.9	59.5	81.9	76.2	62.2
	✓		✓	✓	64.3	56.8	38.3	61.4	54.7	37.1
		✓	✓		47.3	45.3	32.8	56.5	53.2	40.0
		✓	✓	✓	51.0	44.4	27.5	50.0	44.4	28.2

summarizes these results.

Across all test settings, the anonymization quality of the images generated by our framework generally outperforms SecureReID. Specifically, when using blurring to initialize the supervised images on the SYSU-MM01(all search), the SSIM and PSNR [27] for visible images decreased by 0.05 and 1.3, and the SSIM and PSNR for infrared images also dropped by 0.08 and 1.0. Moreover, the SSIM and PSNR values between the anonymized and raw images fall below the predefined thresholds 0.5 and 15, clearly indicating that the anonymized images conceal identity details so effectively that they cannot be distinguished by the naked eye.

4.5. Ablation Studies

Ablation studies are conducted on SYSU-MM01 (using mosaic initialization). We present the key **Anon-to-Raw** setting results in Table 4. Complete results for all settings are in the **supplementary material**.

Baseline As shown in the first block of Table 4, Our baseline model achieves only 47.9% Rank-1 accuracy in the Anon→Raw setting. This severe degradation highlights the baseline’s inability to handle the information loss and domain gaps caused by anonymization, rendering it insufficient for robust PP-VI-ReID.

Effectiveness of KPR. KPR mitigates information loss

Table 4. **Ablation study on SYSU-MM01 in Anon-to-Raw setting under mosaic initialization.** “KPR” refers to Keypoint-Preserving Regularization. “DCMA” means Differential Consistency-guided Modality Alignment. Rank1 accuracy(%), mAP (%) and mINP (%) are reported.

Components			All Search			Indoor Search		
Baseline	KPR	DCMA	Rank1	mAP	mINP	Rank1	mAP	mINP
✓			47.9	47.3	33.5	59.1	66.7	63.6
✓	✓		51.6	51.3	37.6	60.6	68.3	64.8
✓		✓	49.9	48.9	35.0	60.5	68.0	64.6
✓	✓	✓	53.3	51.9	37.9	62.9	70.6	67.6

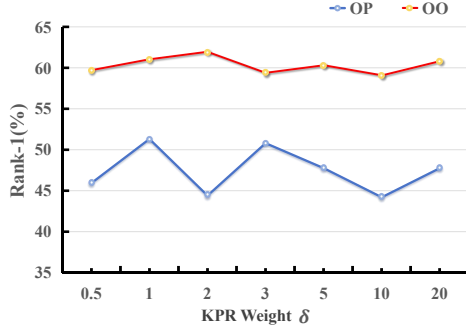


Figure 3. The impact of KPR weight value on SYSU-MM01 dataset. OO refers to raw query/gallery, OP refers to raw query, anonymized gallery.

by preserving the structural integrity of human keypoints. Its impact is most significant on anonymized queries, boosting Rank-1 accuracy by 3.7% in the Anon→Raw setting.

Effectiveness of DCMA. DCMA enhances robustness by enforcing a consistent anonymization effect across modalities. This improves Rank-1 accuracy by 2.0% in the Anon→Raw setting, highlighting its crucial role in learning an anonymization offset for reliable cross-modal retrieval.

Finally, combining both modules yields the best performance, with our full model achieving a peak 53.3% Rank-1 accuracy(5.4% over the baseline) in the Anon→Raw setting. This confirms KPR’s preservation and DCMA’s consistency are complementary and crucial for PP-VI-ReID.

4.6. Further Analysis

Parameter analysis. The influence of hyperparameter δ in Eq.14 is explored. As shown in Figure 3, when $\delta = 1.0$, our framework achieves the best result. More parameter analyses are available in the **supplementary material**.

Visualization. Figure 4 visualizes 20 identities from SYSU-MM01. Compared to the baseline, PPA shows significantly better convergence between raw infrared and anonymized visible positive samples. This demonstrates PPA’s efficacy in mitigating the mixed gaps, although some dispersed samples indicate room for improvement.

Reliability of Pose Estimation on IR Images. To validate the cross-modal pose detection, we applied the frozen RGB-trained YOLOv8n-pose model directly to infrared data

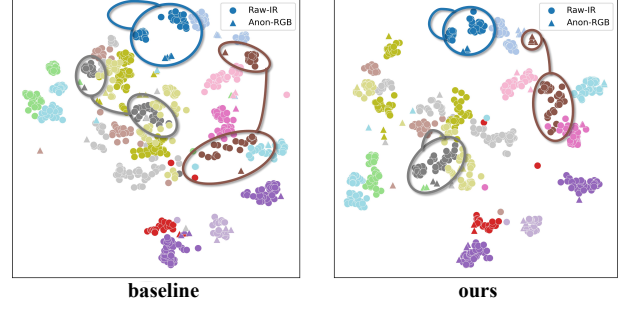


Figure 4. The t-SNE visualization of randomly selected identities. Colors indicate identities. Raw-IR indicates raw infrared image features, and Anon-RGB indicates anonymized visible features.

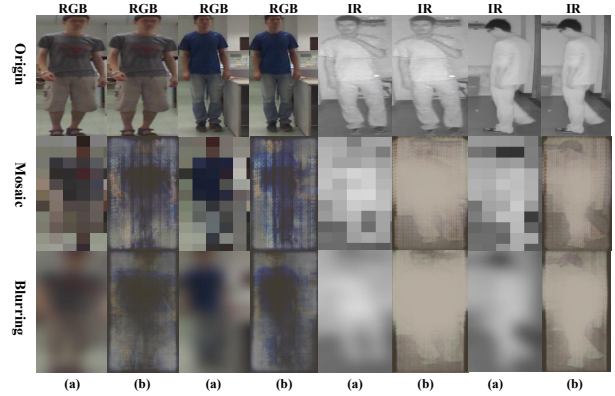


Figure 5. **Qualitative results on privacy protection.** (a) shows algorithm-generated images, while (b) displays the anonymous images generated by our method. Compared to SecureReID, our method preserves identity-critical keypoints and contours, enabling robust cross-modal alignment.

without fine-tuning. Experiments on SYSU-MM01 show the IR performance (95.46% detection rate, 0.763 average confidence) robustly rivals the RGB modality (95.11%, 0.757). This confirms that thermal contours provide sufficient structural cues for reliable keypoint extraction.

5. Conclusion

We introduce Privacy-Preserving Visible-Infrared Person Re-Identification (PP-VI-ReID) for privacy-respecting 24-hour surveillance. We propose the Precise Privacy-preserving and Alignment Network (PPA) addressing anonymization-induced information loss and cross-modal misalignment. By integrating Keypoint-Preserving Regularization (KPR) for structural integrity and Differential Consistency-guided Modality Alignment (DCMA) for robust feature alignment, PPA establishes a strong baseline for PP-VI-ReID task. This work advances practical, privacy-conscious cross-modal ReID systems.

Acknowledgements. This work is partially supported by National Natural Science Foundation of China under Grants (62501428,62225113), Postdoctoral Fellowship Program of CPSF under Grant Number (GZC20241268, 2024M762479), Hubei Provincial Natural Science Foundation of China (2025AFB219), National Natural Science Foundation of China under Grant (T2541022). The supercomputing system at the Supercomputing Center of Wuhan University supported the numerical calculations in this paper.

References

- [1] Jia-Wei Chen, Li-Ju Chen, Chia-Mu Yu, and Chun-Shien Lu. Perceptual indistinguishability-net (pi-net): Facial image obfuscation with manipulable semantics. In *CVPR*, pages 6478–6487, 2021. 2
- [2] Shuyi Chen, Mang Ye, and Bo Du. Rotation invariant transformer for recognizing object in uavs. In *ACM MM*, pages 2565–2574, 2022. 2
- [3] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *CVPR*, pages 248–255, 2009. 6
- [4] Shuguang Dou, Xinyang Jiang, Qingsong Zhao, Dongsheng Li, and Cairong Zhao. Towards privacy-preserving person re-identification via person identify shift. *arXiv preprint arXiv:2207.07311*, 2022. 2
- [5] Oran Gafni, Lior Wolf, and Yaniv Taigman. Live face de-identification in video. In *ICCV*, pages 9378–9387, 2019. 2
- [6] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, pages 770–778, 2016. 6
- [7] Alexander Hermans, Lucas Beyer, and Bastian Leibe. In defense of the triplet loss for person re-identification. *arXiv preprint arXiv:1703.07737*, 2017. 5
- [8] Zhangyi Hu, Bin Yang, and Mang Ye. Empowering visible-infrared person re-identification with large foundation models. *NIPS*, 37:117363–117387, 2024. 2, 3
- [9] Håkon Hukkelås, Rudolf Mester, and Frank Lindseth. Deepprivacy: A generative adversarial network for face anonymization. In *ISVC*, pages 565–578, 2019. 2
- [10] Phillip Isola, Jun-Yan Zhu, Tinghui Zhou, and Alexei A. Efros. Image-to-image translation with conditional adversarial networks. In *CVPR*, 2017. 4
- [11] Kongzhu Jiang, Tianzhu Zhang, Xiang Liu, Bingqiao Qian, Yongdong Zhang, and Feng Wu. Cross-modality transformer for visible-infrared person re-identification. In *ECCV*, pages 480–496, 2022. 3
- [12] Zhenzhong Kuang, Zhiqiang Guo, Jinglong Fang, Jun Yu, Noboru Babaguchi, and Jianping Fan. Unnoticeable synthetic face replacement for image privacy protection. *Neurocomputing*, 457:322–333, 2021. 2
- [13] Zhenzhong Kuang, Huigui Liu, Jun Yu, Aikui Tian, Lei Wang, Jianping Fan, and Noboru Babaguchi. Effective de-identification generative adversarial network for face anonymization. In *ACM MM*, pages 3182–3191, 2021. 2
- [14] Diangang Li, Xing Wei, Xiaopeng Hong, and Yihong Gong. Infrared-visible cross-modal person re-identification with an x modality. In *AAAI*, pages 4610–4617, 2020. 2, 3
- [15] Tao Li and Lei Lin. Anonymousnet: Natural face de-identification with measurable privacy. In *CVPR Workshops*, 2019. 2
- [16] Xiao Li, Yi-Xing Peng, and Wei-Shi Zheng. Protecting feature privacy in person re-identification. *IEEE TPAMI*, 47(11):9637–9654, 2025. 2, 6
- [17] Yutian Lin, Liang Zheng, Zhedong Zheng, Yu Wu, Zhilan Hu, Chenggang Yan, and Yi Yang. Improving person re-identification by attribute and identity learning. *PR*, 95:151–161, 2019. 2
- [18] Hu Lu, Xuezhong Zou, and Pingping Zhang. Learning progressive modality-shared transformers for effective visible-infrared person re-identification. In *AAAI*, pages 1835–1843, 2023. 3
- [19] Maxim Maximov, Ismail Elezi, and Laura Leal-Taixé. Cigan: Conditional identity anonymization generative adversarial networks. In *CVPR*, pages 5447–5456, 2020. 2
- [20] Dat Tien Nguyen, Hyung Gil Hong, Ki Wan Kim, and Kang Ryoung Park. Person recognition system based on a combination of body images from visible light and thermal cameras. *Sensors*, 17(3):605, 2017. 6
- [21] Hyunjong Park, Sanghoon Lee, Junghyup Lee, and Bumsub Ham. Learning by aligning: Visible-infrared person re-identification using cross-modal correspondences. In *CVPR*, pages 12046–12055, 2021. 3
- [22] Hugo Proença. The uu-net: Reversible face de-identification for visual surveillance video footage. *IEEE TCSVT*, 32(2): 496–509, 2021. 2
- [23] Zhigang Su, Dawei Zhou, Nannan Wang, Decheng Liu, Zhen Wang, and Xinbo Gao. Hiding visual information via obfuscating adversarial perturbations. In *ICCV*, pages 4356–4366, 2023. 2, 3
- [24] Qianru Sun, Liqian Ma, Seong Joon Oh, Luc Van Gool, Bernt Schiele, and Mario Fritz. Natural and effective obfuscation by head inpainting. In *CVPR*, pages 5050–5059, 2018. 2
- [25] Qianru Sun, Ayush Tewari, Weipeng Xu, Mario Fritz, Christian Theobalt, and Bernt Schiele. A hybrid model for identity obfuscation by face replacement. In *ECCV*, pages 553–569, 2018. 2
- [26] Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jon Shlens, and Zbigniew Wojna. Rethinking the inception architecture for computer vision. In *CVPR*, 2016. 5
- [27] Zhou Wang, A.C. Bovik, H.R. Sheikh, and E.P. Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE TIP*, 13(4):600–612, 2004. 6, 7
- [28] Ancong Wu, Wei-Shi Zheng, Hong-Xing Yu, Shaogang Gong, and Jianhuang Lai. Rgb-infrared cross-modality person re-identification. In *ICCV*, pages 5380–5389, 2017. 3, 6
- [29] Qiong Wu, Pingyang Dai, Jie Chen, Chia-Wen Lin, Yongjian Wu, Feiyue Huang, Bineng Zhong, and Rongrong Ji. Discover cross-modality nuances for visible-infrared person re-identification. In *CVPR*, pages 4330–4339, 2021. 3

- [30] Yifan Wu, Fan Yang, and Haibin Ling. Privacy-protective-gan for face de-identification. *arXiv preprint arXiv:1806.08906*, 2018. [2](#)
- [31] Zesen Wu and Mang Ye. Unsupervised visible-infrared person re-identification via progressive graph matching and alternate learning. In *CVPR*, pages 9548–9558, 2023. [1](#), [3](#)
- [32] Kunlun Xu, Xu Zou, and Jiahuan Zhou. Lstkc: Long short-term knowledge consolidation for lifelong person re-identification. In *AAAI*, pages 16202–16210, 2024. [3](#)
- [33] Bin Yang, Mang Ye, Jun Chen, and Zesen Wu. Augmented dual-contrastive aggregation learning for unsupervised visible-infrared person re-identification. In *ACM MM*, pages 2843–2851, 2022. [1](#), [2](#), [3](#)
- [34] Bin Yang, Jun Chen, Cuiqun Chen, and Mang Ye. Dual consistency-constrained learning for unsupervised visible-infrared person re-identification. *IEEE TIFS*, 19:1767–1779, 2023. [1](#)
- [35] Bin Yang, Jun Chen, and Mang Ye. Towards grand unified representation learning for unsupervised visible-infrared person re-identification. In *ICCV*, pages 11069–11079, 2023. [3](#)
- [36] Bin Yang, Jun Chen, and Mang Ye. Shallow-deep collaborative learning for unsupervised visible-infrared person re-identification. In *CVPR*, pages 16870–16879, 2024. [2](#), [3](#)
- [37] Mouxing Yang, Zhenyu Huang, Peng Hu, Taihao Li, Jiancheng Lv, and Xi Peng. Learning with twin noisy labels for visible-infrared person re-identification. In *CVPR*, pages 14308–14317, 2022. [3](#)
- [38] Haoyu Yao, Bin Yang, Wenke Huang, Bo Du, and Mang Ye. Unsupervised visible-infrared person re-identification under unpaired settings. In *ICCV*, pages 11916–11926, 2025. [3](#)
- [39] Mang Ye, Zheng Wang, Xiangyuan Lan, and Pong C Yuen. Visible thermal person re-identification via dual-constrained top-ranking. In *IJCAI*, pages 1092–1099, 2018. [3](#)
- [40] Mang Ye, Xiangyuan Lan, Zheng Wang, and Pong C Yuen. Bi-directional center-constrained top-ranking for visible thermal person re-identification. *IEEE TIFS*, 15:407–419, 2019.
- [41] Mang Ye, Jianbing Shen, David J. Crandall, Ling Shao, and Jiebo Luo. Dynamic dual-attentive aggregation learning for visible-infrared person re-identification. In *ECCV*, pages 229–247, 2020. [3](#)
- [42] Mang Ye, Jianbing Shen, Gaojie Lin, Tao Xiang, Ling Shao, and Steven CH Hoi. Deep learning for person re-identification: A survey and outlook. *IEEE TPAMI*, 44(6): 2872–2893, 2021. [1](#), [2](#), [3](#), [4](#), [6](#)
- [43] Mang Ye, Zesen Wu, Cuiqun Chen, and Bo Du. Channel augmentation for visible-infrared re-identification. *IEEE TPAMI*, 46(4):2299–2315, 2023. [2](#), [3](#), [6](#)
- [44] Mang Ye, Wei Shen, Junwu Zhang, Yao Yang, and Bo Du. Securereid: Privacy-preserving anonymization for person re-identification. *IEEE TIFS*, 19:2840–2853, 2024. [2](#), [6](#), [7](#)
- [45] Mang Ye, Shuoyi Chen, Chenyue Li, Wei-Shi Zheng, David Crandall, and Bo Du. Transformer for object re-identification: A survey. *IJCV*, 133(5):2410–2440, 2025. [2](#)
- [46] Mang Ye, Zesen Wu, and Bo Du. Dual-level matching with outlier filtering for unsupervised visible-infrared person re-identification. *IEEE TPAMI*, 47(5):3815–3829, 2025. [3](#)
- [47] Zhengxin You, Sheng Li, Zhenxing Qian, and Xinpeng Zhang. Reversible privacy-preserving recognition. In *ICME*, pages 1–6, 2021. [2](#)
- [48] Can Zhang, Hong Liu, Wei Guo, and Mang Ye. Multi-scale cascading network with compact feature learning for rgb-infrared person re-identification. In *ICPR*, pages 8679–8686, 2021. [3](#)
- [49] Delong Zhang, Yi-Xing Peng, Xiao-Ming Wu, Ancong Wu, and Wei-Shi Zheng. Pixelfade: Privacy-preserving person re-identification with noise-guided progressive replacement. In *ACM MM*, pages 6326–6334, 2024. [2](#), [3](#)
- [50] Yukang Zhang, Yan Yan, Yang Lu, and Hanzi Wang. Adaptive middle modality alignment learning for visible-infrared person re-identification. *IJCV*, 133(4):2176–2196, 2025. [2](#)
- [51] Liang Zheng, Liyue Shen, Lu Tian, Shengjin Wang, Jingdong Wang, and Qi Tian. Scalable person re-identification: A benchmark. In *ICCV*, pages 1116–1124, 2015. [6](#)
- [52] Liang Zheng, Yi Yang, and Alexander G. Hauptmann. Person re-identification: Past, present and future. *arXiv preprint arXiv:1610.02984*, 2016. [2](#)
- [53] Xiangtao Zheng, Xiumei Chen, and Xiaoqiang Lu. Visible-infrared person re-identification via partially interactive collaboration. *IEEE TIP*, 31:6951–6963, 2022. [3](#)
- [54] Xiyang Zheng, Yukang Zhang, Yang Lu, and Hanzi Wang. Semi-supervised visible-infrared person re-identification via modality unification and confidence guidance. In *ACM MM*, pages 5761–5770, 2024. [3](#)